PROJECT III (CACS452) | BCA EIGHTH SEMESTER

Anka

Student Result Analysis and Ranking System

Results, back papers, ranking, insights and daily classes for institutions of any level

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The problem

- Marks are copied by hand between registers and spreadsheets, so errors reach students.
- Back papers are hard to track and their effect on GPA and CGPA is unclear.
- Each program has different grading rules, but most tools allow only one.
- Printed marksheets cannot be verified.
- Weak students are noticed after the final exam, not before it.
- Class timetables, diaries and assignments live in notebooks and chat groups.
- Local software is often tied to one institution's name or rules.

Objectives

Calculate

Configurable grading engine: TU GPA, NEB grade points, divisions; SGPA and CGPA

Track

Back papers as separate sittings with full history

Rank and analyze

Merge sort ranking, heap, statistics, regression, correlation, k-means

Protect

Role based access, QR verified marksheets, explainable risk score

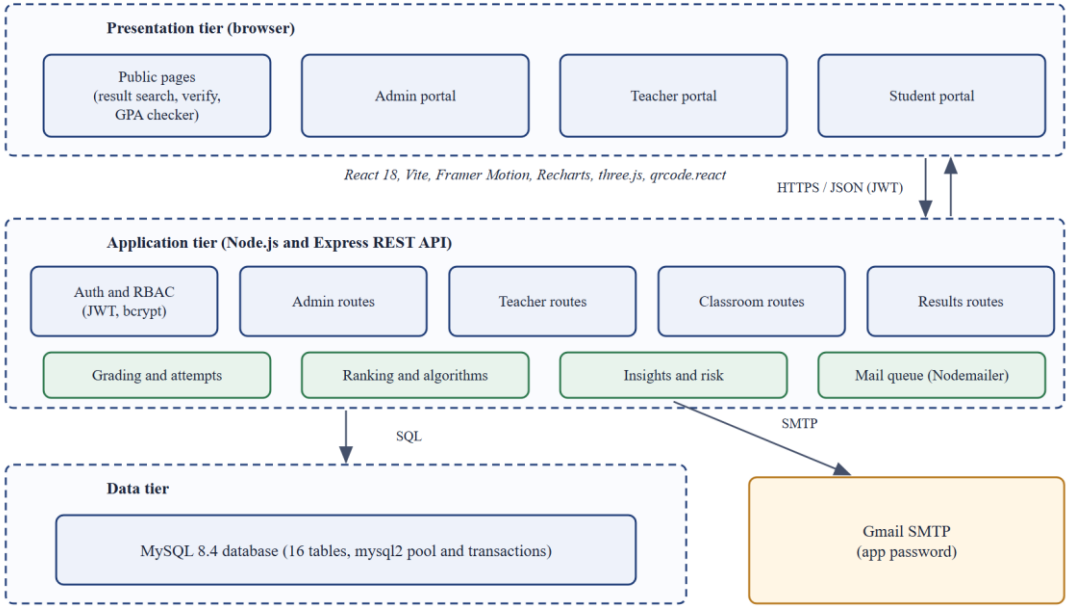
Communicate

Email queue with retries; notices; audit log

Run the class

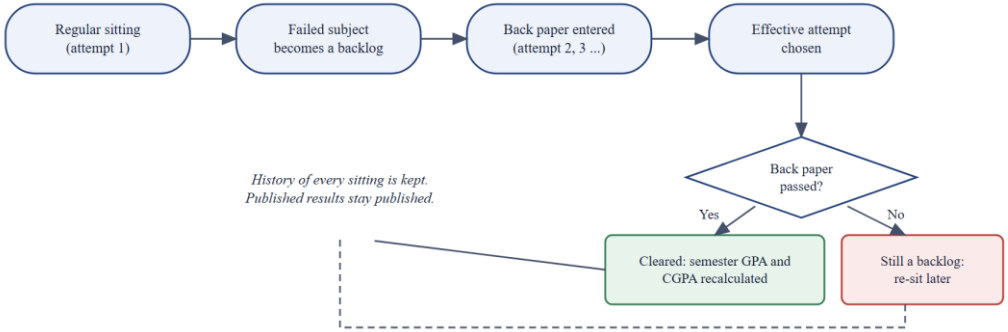
Timetable, class diary, assignments; unbranded for any institution

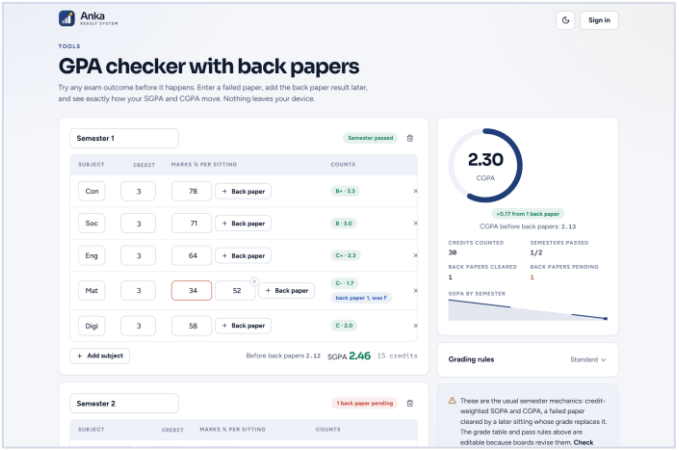
System architecture



- React 18 client with a shared design system
- Express REST API: 82 endpoints, 5 routers
- MySQL 8.4: 16 tables, transactions, foreign keys
- Gmail SMTP through a retrying queue
- JWT sessions, bcrypt, role checks, rate limits

Grading engine and back papers



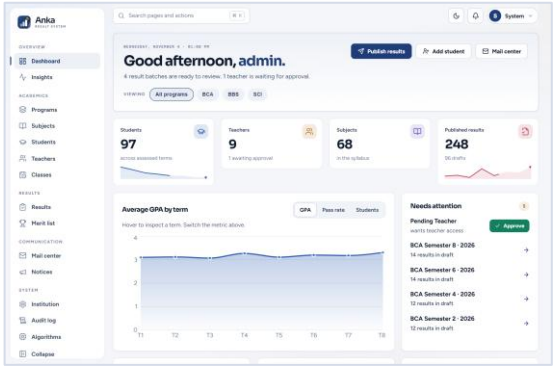


- SGPA = sum(point x credit) / sum(credit); CGPA over semesters
- Pass needs the pass mark overall and in the external part
- Grade lookup by binary search over editable bands
- Back paper = new attempt row; effective attempt chosen; semester result recalculated; history kept

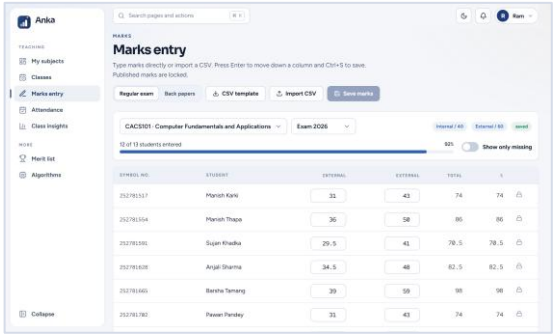
Algorithms implemented from scratch

Algorithm	Used for	Cost
Binary search	Grade band, percentile	O(log n)
Merge sort + competition rank	Merit list, class rank with ties	O(n log n)
Binary min-heap	Top performers (top-k)	O(n log k)
Statistics, z-score	Mean, median, deviation, pass rate	O(n log n)
Linear regression	GPA trend and forecast	O(n)
Pearson correlation	Attendance vs GPA	O(n)
K-means (farthest-first start)	Support groups	O(n k i)
Weighted risk score	Explainable early warning	O(1)

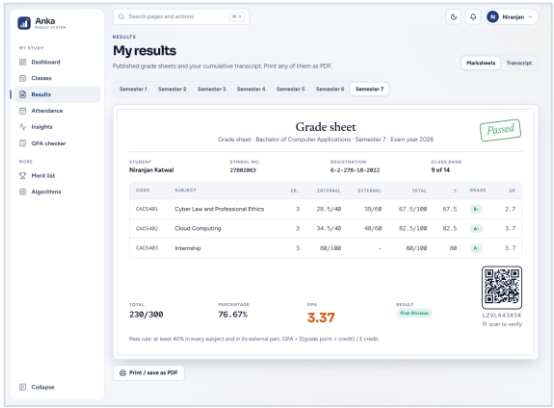
The system at work



Administrator dashboard



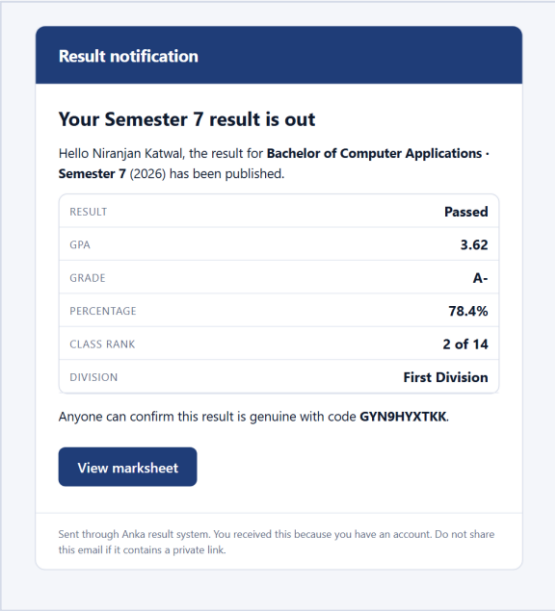
Marks entry grid



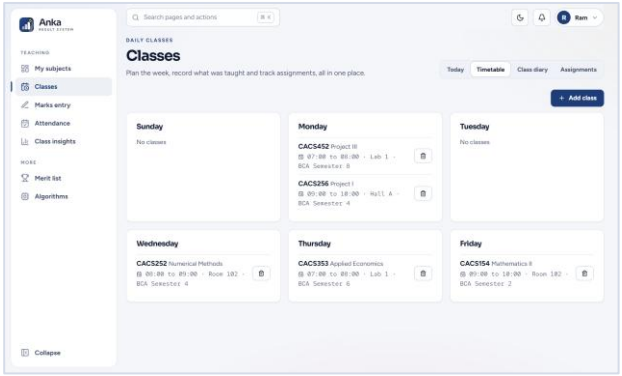
Marksheet with QR code

Neutral navy design system, light and dark themes, command palette, responsive down to phone width.

Email and daily classes



Result email as received (rendered)



- Eight HTML templates in the web palette
- Queue, 3 attempts, full log and preview
- Timetable rejects teacher and class clashes
- Class diary and assignment tracking

Testing and results

18	58	43	32
unit tests	API flow checks	smoke checks	classroom checks

- Browser end-to-end suite drives the real pages: 56 of 58 steps passed on the best clean run
- It found real defects: command palette stuck open, two screens with a missing import
- 59 endpoints x 4 roles permission matrix; email tested against a local SMTP server
- Grade table and back paper rules are editable: to be confirmed against the official TU regulation

Conclusion and future work

Delivered

- Configurable grading with back papers
- Ranking, analytics, risk score
- QR verified marksheets, email queue
- Timetable, diary, assignments
- Unbranded: any institution

Next

- Confirm rules with the official TU regulation
- Several institutions on one server
- Mobile app and SMS
- File upload for assignments
- Signed PDF marksheets

Thank you. Questions? github.com/itsniranjannn/StuResSystem